

Reconstructing point sets from unlabeled distance measurements

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Rigidity Theory: Structures, Data, and Control

Lancaster University, September 15, 2026

Fix an integer $d \geq 1$ and a graph G . I will assume:

$$|V(G)| \geq d + 2, \text{ and } G \text{ has no isolated vertices.}$$

For each $p : V(G) \rightarrow \mathbb{R}^d$, we let

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- $M_{d,G} = \mathbb{C}^{E(G)} \Leftrightarrow G$ is $\mathcal{R}_d$ -independent,
- For the cycle graph C_n , we have

$$M_{1,C_n} = \{g_n = 0\} \subseteq \mathbb{C}^n,$$

where

$$g_n(x_1, \dots, x_n) = \prod (\sqrt{x_1} \pm \sqrt{x_2} \pm \dots \pm \sqrt{x_n}).$$

What is encoded by the variety?

$M_{d,G}$ uniquely determines the generic rigidity matroid $\mathcal{R}_d(G)$:

$$I \subseteq E(G) \text{ independent} \Leftrightarrow \pi_I : M_{d,G} \rightarrow \mathbb{C}^I \text{ is dominant}$$

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$M_{d,G}$ and $n = |V(G)|$ tells us whether:

- G is d -rigid $\Leftrightarrow \dim(M_{d,G}) = dn - \binom{d+1}{2}$
- G is globally d -rigid $\Leftrightarrow d$ -rigid and $\delta(M_{d,G}) = \binom{d+1}{2}$

Main question (roughly)

Which graphs G are uniquely determined by $M_{d,G}$?

“ G can be reconstructed from unlabeled distance measurements”

Many variants: weakly/strongly/fully reconstructible graphs

$\implies$ I will focus on **full reconstructibility**.

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Notation

We write $M_{d,G} =_{\psi} M_{d,H}$ if $M_{d,H} = \Psi(M_{d,G})$.

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..and another one

We write $G \cong_{\psi} H$ if there exists $\varphi : V(G) \rightarrow V(H)$ s.t.
 $\psi(uv) = \varphi(u)\varphi(v)$ for all $uv \in E(G)$.

(= “ ψ is induced by a graph isomorphism”)

Full reconstructibility

Definition

We say that G is **fully reconstructible** in $\mathbb{C}^d$ if for any graph H and bijection $\psi : E(G) \rightarrow E(H)$,

$$M_{d,G} =_{\psi} M_{d,H} \implies G \cong_{\psi} H.$$

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Necessary conditions

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For a graph G , fully reconstructible in $\mathbb{C}^d \Rightarrow 3$ -connected.

Proof: Whitney switch.

One-dimensional reconstructibility

Theorem (G.-Jordán 2021)

G is fully reconstructible in $\mathbb{C}^1$ iff it is 3-connected.

Necessity: previous slide.

Sufficiency: $M_{1,G} =_{\psi} M_{1,H}$ implies $\mathcal{R}_1(G) \cong_{\psi} \mathcal{R}_1(H)$.

By a theorem of Whitney (from 1933!), $G \cong_{\psi} H$.

Theorem (Gortler-Theran-Thurston 2019)

Let $d \geq 2$, and suppose that G is globally d -rigid.

If $M_{d,G} =_{\psi} M_{d,H}$ and $|V(G)| = |V(H)|$, then $G \cong_{\psi} H$.

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Proof idea:

- Show that $M_{d-1,G} =_{\psi} M_{d-1,H}$.
- By iterating, $M_{1,G} =_{\psi} M_{1,H}$.
- G is 3-connected $\Rightarrow$ fully rec. in $\mathbb{C}^1 \Rightarrow G \cong_{\psi} H$.

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- Use magic argument to deduce $|V(H)| = |V(G)|$.
- Apply the Gortler-Theran-Thurston theorem.

Two-dimensional reconstructibility

Theorem (G.-Jordán 2021)

G is fully reconstructible in $\mathbb{C}^1$ iff it is 3-connected.

Theorem (G.-Jordán 2021)

G is fully reconstructible in $\mathbb{C}^2$ iff it is globally 2-rigid.

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G is fully reconstructible in $\mathbb{C}^2$ iff it is globally 2-rigid.

For full reconstructibility in $\mathbb{C}^d$, $d \geq 3$:

- no combinatorial characterization is known,
- being globally d -rigid is sufficient, but not necessary.

An interesting example

Theorem (Bernstein-Gortler 2022)

$K_{5,5}$ is fully reconstructible in $\mathbb{C}^3$.

Their proof:

- Geometric idea: in every generic realization $(K_{5,5}, p)$ in $\mathbb{R}^3$, every equilibrium stress is **balanced** (sums to zero).
- This is encoded in the measurement variety.
- Show (computationally!) that no other $\mathcal{R}_3$ -circuit on 25 edges has this property.

Reconstructibility of complete bipartite graphs

Theorem (Bernstein-Gortler 2022)

$K_{5,5}$ is fully reconstructible in $\mathbb{C}^3$.

Theorem (G. 2026+)

$K_{k,\ell}$ is fully reconstructible in $\mathbb{C}^d$ for all $k, \ell \geq d + 2$.

Proof idea: look at the **generic stress matroid** of $K_{k,\ell}$.

Conjecture

Let G be a graph with $M_{d,G} \neq \mathbb{C}^{E(G)}$. Then for any graph H , integer $d' \geq 1$, and bijection $\psi : E(G) \rightarrow E(H)$,

$$M_{d,G} =_{\psi} M_{d',H} \implies d = d'.$$

“ $M_{d,G}$ uniquely determines d .”

I can show this for

- $d = 1, 2$,
- $d = 3$ and G globally 3-rigid,
- d arbitrary and G highly vertex-connected (I think).

Thank you!

Papers:

- Gortler, Theran, Thurston, **Generic unlabeled global rigidity**, *Forum of Mathematics, Sigma*, 2019.
- Garamvölgyi, Gortler, Jordán, **Globally rigid graphs are fully reconstructible**, *Forum of Mathematics, Sigma*, 2022.
- Bernstein, Gortler, **$K_{5,5}$ is fully reconstructible in $\mathbb{C}^3$** , *Discrete Applied Mathematics*, 2022.
- A preprint by me, soon. . .